

DATA SCIENCE RESOURCES

Healthcare Projects

Project 1. Patient Risk Scoring System

Problem Statement: Hospitals need to identify high-risk patients early to prevent readmissions, improve outcomes, and allocate ICU resources efficiently.

Objective: Build a patient risk scoring system to classify patients into high, medium, or low-risk categories based on vitals, lab results, and medical history.

Dataset Source

- [MIMIC-III or MIMIC-IV \(MIT PhysioNet\)](#)
- [Heart Disease UCI Dataset](#)

Techniques & Concepts

- Classification (Logistic Regression, XGBoost, SVM)
- Feature importance using SHAP values
- Risk bucket segmentation
- ROC-AUC, recall, and precision evaluation

Tools & Libraries: Python, Scikit-learn, SHAP, Pandas, Seaborn, Streamlit

Expected Outcome: A system that outputs a risk score for patients and flags high-risk individuals for proactive intervention.

Project 2. Disease Diagnosis Using Symptoms

Problem Statement: General practitioners often receive vague symptoms, making early disease identification difficult without clinical tests.

Objective: Use machine learning to predict possible diseases based on symptom inputs provided by patients or wearable devices.

Dataset Source

- [Disease Symptom Dataset](#)
- [Simulated clinical symptom questionnaires](#)

Techniques & Concepts

- Multi-class classification (Naive Bayes, Random Forest, Neural Nets)
- Text vectorization of symptoms (TF-IDF, One-hot encoding)
- Symptom clustering
- Evaluation via confusion matrix, precision, and recall

Tools & Libraries: Python, Pandas, Scikit-learn, NLTK, Streamlit

Expected Outcome: A model that provides a top-k ranked list of likely diseases based on user-entered symptoms, useful for triage and preliminary diagnosis.

Project 3. Medical Image Segmentation (Tumor/Organ Detection)

Problem Statement: Manual image annotation by radiologists is time-consuming and prone to subjective interpretation. Accurate segmentation is critical for diagnostics and surgery planning.

Objective: Use deep learning to segment medical images and highlight areas of concern such as tumors, lungs, or other organs.

Dataset Source

- [Brain Tumor MRI Dataset](#)
- [LUNA16 CT Scans](#)

Techniques & Concepts

- Convolutional Neural Networks (CNNs)
- U-Net, SegNet, or DeepLabV3+ architecture
- Dice coefficient, IoU (Intersection over Union)
- Preprocessing: resizing, normalization, augmentation

Tools & Libraries: Python, TensorFlow/Keras or PyTorch, OpenCV, NumPy, Matplotlib

Expected Outcome: A fully trained model that segments critical regions in MRI, CT, or dermoscopic images and can be integrated into a clinical pipeline.
